

Why were 18 of 30 orders late?

Monday Orders (30 total)



Order Process (timings unknown)



Scanner Charging Checklist

Date	Scanner 1	Scanner 2
Mon	☐	☐
Tue	☐	☐
Wed	☐	☐

What problem are we solving?

3 hours: 50-minute lesson, 10-minute break, 50-minute guided practice, 10-minute break, 50-minute applied practice, and 10-minute wrap-up.

SITUATION

The problem

A small warehouse shipped 18 of 30 orders late on Monday. Staff report that two barcode scanners ran out of battery, but the manager does not yet know whether this explains every delay.

TARGET

Learning goal

Map a simple process, distinguish evidence from assumptions, and propose a small test of a likely process cause.

Practice material: Use the practice material supplied in the session Markdown.

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Practice facts

- Orders are picked, scanned, packed, then handed to a courier. Two scanners were uncharged at opening. The charging checklist was blank for three days. No timing data by step is available.

BOUNDARY

Use the supplied facts

A small warehouse shipped 18 of 30 orders late on Monday. Staff report that two barcode scanners ran out of battery, but the manager does not yet know whether this explains every delay.

Concepts for this problem

Keep each concept tied to the practical decision.

UNDERSTAND

Core concepts

- A bottleneck is the step that currently limits flow; find it with timing or queue evidence, not a guess.
- The 5 Whys asks why a problem occurred repeatedly, but each answer is a hypothesis until checked.
- Focus on process conditions and safeguards, not blame. One observation may not explain all late orders.

APPLY

A simple method

- State the measurable problem; build a short cause chain and label evidence versus unknowns; choose a low-risk test and a measure.

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

Who caused all these late orders?

AVOID

Weak example 2

Do five whys and tell me the root cause.

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

Use only the facts below. List what is observed, what is unknown, and up to three possible causes of late orders. Mark each cause as a hypothesis; do not assign blame.

USE

Improved example 2

For the hypothesis that uncharged scanners slowed picking, propose a one-week test. Specify the daily measure, comparison, and a result that would weaken this hypothesis. Do not assume it explains all 18 late orders.

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Ask AI to list evidence, unknowns, and possible causes. Draw a simple chain from uncharged scanners to a possible picking delay. Mark every unverified link with "to check."

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- Draft a one-week opening checklist test. Record scanner charge status and late orders daily; compare with the previous week if reliable records exist. State why the result can suggest but not prove causation.

How to know the work is complete

Check the output before sharing or acting on it.

SUBMIT

Deliverable

- A one-page cause map and a test plan with a measure, time period, and limitation.

VERIFY

Success check

- The response separates observations from hypotheses, avoids blaming workers, and defines a measurable test.